

this printer. This technique is called 'Fused Filament Fabrication' since the additive source is in the form of spooled filaments of plastic.

Printers of this design are capable of employing nearly any material that has a creamy viscosity, including materials such as clay, silicone, chocolate, cheese, frosting, cement and certain metals. The heating necessary for different additives is either done through the ejection nozzle or the additive storage unit based on the required melting point.

4. Selective Laser Sintering (SLS)

The fourth method uses powdered materials that are fused together using either heat or adhesives between layers to achieve the desired 3D shape. This method is called 'Selective Laser Sintering' (SLS) and is actually a combination of traditional 3D printing and powered lasers, (instead of UV light). By augmenting the stereolithographic method, SLS replaces the chemical pool with powdered base material and the UV light with a powered laser.

Combining both the methods makes it possible for SLS printers to use not only all plastics but also ceramics and metals to fabricate objects. It has proven to be a cost-effective alternative to other 3D printers in specialised cases which call for the use of materials such as polystyrene, nylon, glass, metals and other exotic additives. The powdered material from these sources is easy to fuse using a laser and once the 3D printing project is completed, the surplus material is left available for reuse. It also removes the need for the use of supports and makes the overall process much more efficient.

Developing technologies

Even as we discuss the world of 3D printer technologies progress is being made in more versatile and innovative methods. Three of the most promising technologies that are likely to find acceptance in the near future are Selective Laser Melting, Electron Beam Melting and Laminated Object Manufacturing.

1. **Selective Laser Melting:** is similar to Selective Laser Sintering but far more advanced as it uses the power of lasers to fully melt the powder granules into a fixed solid layer instead of simply fusing the powder together. This results in stronger and longer lasting objects that can be intensively used. It is similar to the Electron Beam Melting method as well where electron beams are used instead of a UV laser.
2. **The Electron Beam Melting method:** is used for extremely high precision object printing, which is required to make biological grade implants, such as those used in orthopedic surgery. The complete